

🔥 DAY 23 — MULTI-AGENT (SPECIALIZED AGENTS)

✅ GOAL

Upgrade from: One agent ❌

to: Multiple specialized agents ✅

✅ WHAT YOU BUILD

Main Agent (controller)

↓

Sub-agents:

└─ RAG Agent

└─ Excel Agent (already exists)

└─ PDF Table Agent

✅ ✅ STEP 1 — CREATE RAG AGENT

📄 **NEW FILE:** app/rag_agent.py

✅ ADD CODE

```
from models.llm import generate
```

```
class RAGAgent:
```

```
    def init(self, rag_system):
```

```
        self.rag = rag_system
```

```
    def run(self, query):
```

```
        docs = self.rag.search(query)
```

```
        context = "\n".join(docs[:3])
```

```
        prompt = f"""
```

```
Answer using context:
```

```
{context}
```

```
Question:
```

```
{query}
```

```
"""
```

```
    return generate(prompt)
```

✅✅ **STEP 2 — UPDATE MAIN AGENT (COMPONENT SPLIT):** 📄 app/agent.py

🔧 **FIND IMPORTS:** from models.llm import generate

✅ **ADD BELOW:** from app.rag_agent import RAGAgent

✅ **MODIFY __init__**

🔧 **FIND:** self.rag = rag_system

✅ **REPLACE WITH:** self.rag_agent = RAGAgent(rag_system)

✅✅ **STEP 3 — UPDATE TOOL EXECUTION**

🔧 **FIND THIS BLOCK**

```
elif tool == "RAG":
```

```
    docs = self.rag.search(task)
```

```
    result = "\n".join(docs[:3])
```

✅ **REPLACE WITH**

```
elif tool == "RAG":
```

```
    result = self.rag_agent.run(task)
```

✅✅ **STEP 4 — CLEANUP (OPTIONAL BUT GOOD)**

👉 You can remove direct usage of: self.rag

✅✅ FINAL FLOW NOW

User Query

↓

Main Agent (planner)

↓

Delegates to:

└─ RAGAgent ✅

└─ ExcelAgent ✅

└─ PDFTableAgent ✅

↓

Collect results

↓

Final Answer

✅✅ TEST

Run: `uvicorn app.api:app --reload`

✅ Try:

>> summarize document

>> average sales

>> compare trends

✅✅ EXPECTED CHANGE

Before: Single agent logic

After: Modular agents ✅

Better separation ✅

Cleaner architecture ✅

✅✅ WHAT YOU BUILT: Multi-agent architecture

✓ ✓ SYSTEM LEVEL NOW

- ✓ AI system
 - ✓ RAG + Excel + PDF agents
 - ✓ Planner
 - ✓ Memory
 - ✓ Evaluation
 - ✓ Monitoring
 - ✓ ✓ Multi-agent system (NEW)
-

✓ ✓ WHY THIS IS BIG

Now you can:

- add new agents easily
 - scale system
 - debug each agent separately
 - mimic real Copilot architecture
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✓ ✓ INDUSTRY TERM

Multi-agent orchestration

Used in:

- LangGraph
 - AutoGPT
 - Copilot systems
-

✓ ✓ NEXT: Day 24 → Self-correcting agent (retry + fix failures)

If ready: 👉 say **Day 24**